

Modelling Particle Production in Analog Expanding Universes with High-Performance Computing

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Introduction

- ① Big Bang caused particles to be created out of nothing
- ② Can't measure these now
- ③ The behavior of certain quantum systems matches the behavior of these particles
- ④ Can make these quantum systems in the lab $\implies$ can *effectively* observe Big Bang particles
- ⑤ Lab based system = “analog universe”

Related Work

Previous research has been done investigating a variety of different systems

- ① “Hyper-phononic” quasiparticles and Lorentz Violation
- ② Analog massive particles
- ③ Analog particles with non-zero quantum spin
- ④ Analog universes with different expansion behavior
- ⑤ Analog universes with non-zero background velocity
- ⑥ Experimental realizations (some extremely relevant and recent)

This Work

My focus is on

- ① $(3+1)$ D analog universe with high-performance computing
- ② linear dispersion & phononic quasiparticle
- ③ analog massless particles
- ④ analog spin-0 particles
- ⑤ no background velocity

Theory - Speed & Expansion

- Speed of sound in quantum system = Speed of light in analog universe
- Speed decrease & static volume $\iff$ constant speed & expanding volume
- Example: $t = d/v$
 - Distance to grocery store doubles $\implies$ travel time doubles
 - Speed to the grocery store halves $\implies$ travel time doubles

Theory - The Field Equation

- The field equation describes the behavior of
 - a gas of quantum particles made in the lab
 - particles produced just after the Big Bang
- For each 3D wave vector $\mathbf{k}$ such that $||\mathbf{k}|| < k_c$
- Universe expansion starts and stops at finite times

The Field Equation

$$\partial_\eta^2 \tilde{\theta}_{\mathbf{k}} - \frac{1}{2} \frac{\dot{a}(\eta)}{a(\eta)} \partial_\eta \tilde{\theta}_{\mathbf{k}} - c_0^2 k^2 \tilde{\theta}_{\mathbf{k}} = 0 \quad (1a)$$

$$\tilde{\theta}_{\mathbf{k}}(0) = \tilde{\theta}_{\mathbf{k}0} \quad \partial_\eta \tilde{\theta}_{\mathbf{k}}(0) = -\frac{U_0}{\hbar} n_{\mathbf{k}} \quad (1b)$$

- Conserve energy $\implies$ Use stable algorithm
- a_m, b_m are defined numerically for optimality
- $\frac{\text{5th order McAte5 error}}{\text{4th order Ruth (1990) error}} \approx 3\%$

McAte5 Symplectic Integrator

$$1 \leq i \leq 5$$

$$\mathbf{p}_i = \mathbf{p}_0 + k \sum_{m=1}^i b_m \mathbf{F}(\mathbf{q}_{m-1}) \quad (2)$$

$$\mathbf{q}_i = \mathbf{q}_0 + k \sum_{m=1}^i a_m \mathbf{P}(\mathbf{p}_m) \quad (3)$$

Theory - High-Performance Computing

- Field equation for each $k < k_c$
- Use Julia to solve in parallel with multiple GPUs
- Use the CWU “Turing” supercomputer
- 4x4 NVIDIA P100 GPUs - Each 83% performance of modern RTX 3060
- 180 GBps GPU-GPU bandwidth
- 80 GBps (NVLink) GPU-CPU bandwidth

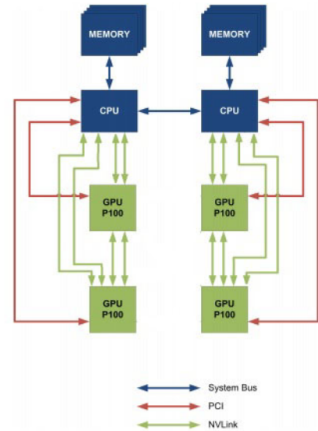


Figure: CPU to GPU and GPU to GPU interconnect using NVLink

Analytic Results - de Sitter Expansion

de Sitter expansion: $a(t) = e^{-t/t_s}$

- Use Mathematica for simple analytic results
- Population distribution highly depends on length of expansion
- Population is relatively high for low k
- Population drops off quickly with k

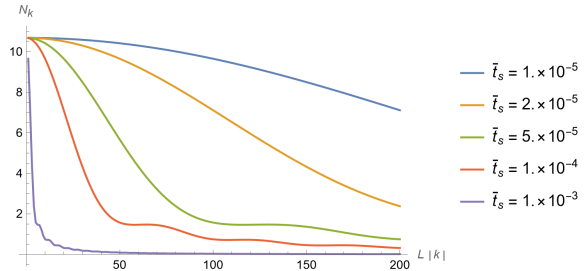


Figure: The momentum distribution when the universe stops expanding for multiple effective rates of expansion

Analytic Results - de Sitter Expansion

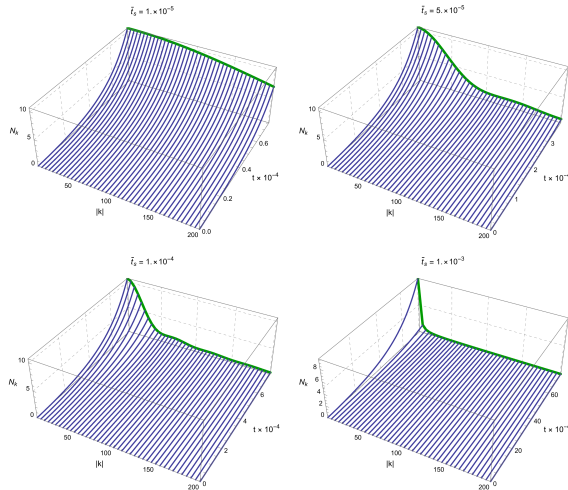


Figure: Momentum distribution throughout the expansion

Discussion - Why High-Performance Computing?

- Analytic methods $\implies$ simple expansions
- Numeric methods $\implies$ complex expansions
- Complex expansions $\implies$ High-performance computational model

Conclusion

- A lot of work has been done into the theory of analog universes for different systems
- A computational model is needed to provide guidance and support to both theory and experiment
- The behavior of particles created from the expansion of the universe is the same as particles in the lab
- Need to find the behavior for many different cases
- Need to support a general, complex expansion
- Use high-performance techniques and hardware for reasonable execution time

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Thank you & Code source

Thank you for your attention.

The code for this project is available at:

- GitHub (QR code): github.com/nondairyneutrino/thesis
- My website: nathanwchapman.com



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